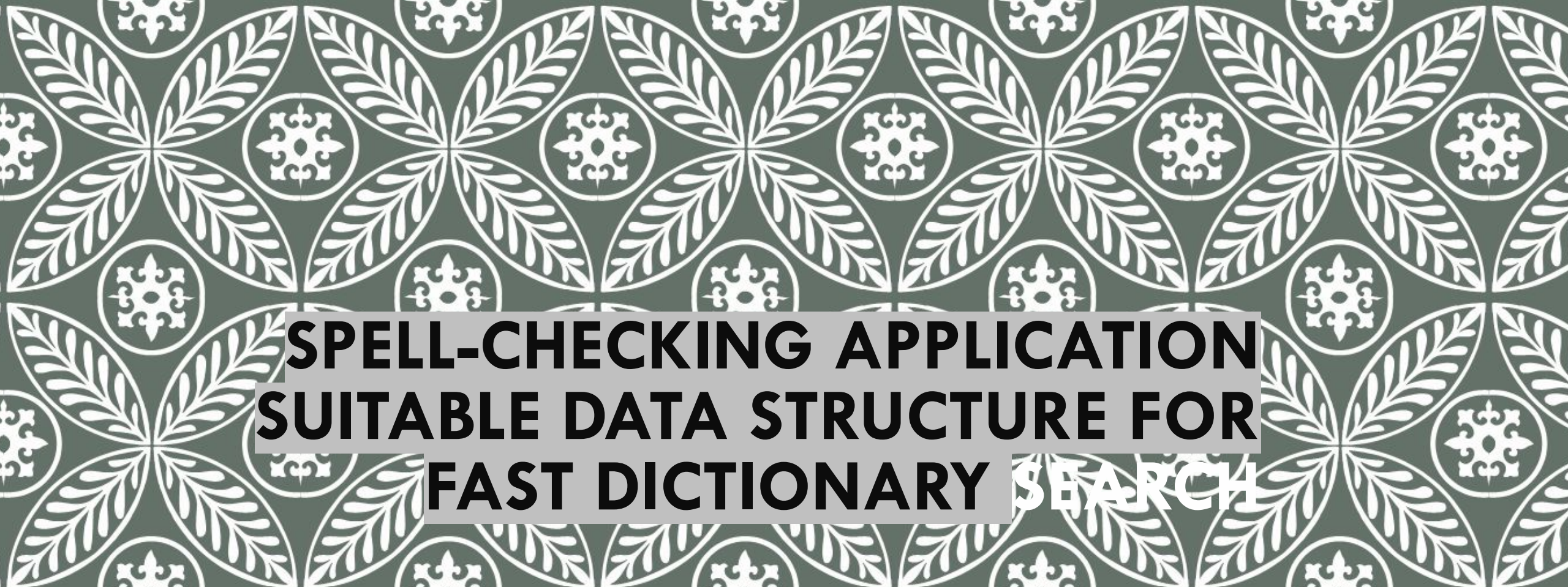




SPELL-CHECKING APPLICATION SUITABLE DATA STRUCTURE FOR FAST DICTIONARY SEARCH

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PROBLEM STATEMENT

Scenario

A spell-checking application verifies whether a typed word exists in its dictionary.

The dictionary contains thousands or millions of words.

The system requires **very fast searching** to provide instant suggestions.

Requirement

Choose a data structure that offers the fastest search speed.

RECOMMENDED DATA STRUCTURE

Data Structure: Hash Table (Hash Map)

Stores words using a **hash function**.

Enables direct access to words.

Lookup is extremely fast.

Widely used in spell-checkers and dictionaries.

ADVANTAGES

Very fast word lookup.

Efficient for large dictionaries.

Supports quick spell verification.

Reduces search time significantly..

CONCLUSION

Recommended Data Structure: Hash Table

Provides **$O(1)$ average search time**, making it ideal for spell-checking applications.

Delivers fast and efficient word verification

Improving user experience.

THANK YOU